LLM-powered scientific discussion generation

Background

The Discussion section of a scientific manuscript is a critical component where authors interpret findings, contextualize results within prior work, and outline implications. Automating the generation of this section could assist researchers in drafting high-quality manuscripts and improve scientific writing support systems. Recent advancements in large language models (LLMs) demonstrate strong capabilities in natural language generation, but their application to structured scientific writing tasks – particularly in generating contextually grounded Discussions based on cited literature – remains underexplored.

Project Description

This project investigates how LLMs can be leveraged to generate Discussion text for a scientific manuscript given a set of cited papers. The task involves producing a Discussion span d' that approximates or improves upon the ground-truth Discussion text d. Inputs include the full text, title, abstract, and metadata of the citing manuscript (excluding its Discussion section) and the cited papers. The dataset for this task has three levels of granularity:

1) **Sentence-level**: Generate a single citing sentences in the Discussion.
2) **Paragraph-level**: Generate a full paragraph in the Discussion.
3) **Section-level**: Generate the entire Discussion section.

Methodology

- **Model benchmarking**: Evaluate different LLMs (open-source and proprietary) under varying prompting strategies (zero-shot, few-shot).
- **Prompt design**: Experiment with prompt engineering techniques (e.g., instruction-tuning prompts, chain-of-thought, context reordering) and assess their impact on output quality.
- **Few-shot selection strategies**: Explore approaches for choosing the most effective examples in few-shot settings.
- **Dataset utilization**: Conduct experiments across sentence-, paragraph-, and section-level datasets to assess performance at different granularities.
- **Evaluation:** Automatic metrics (e.g., BLEU, ROUGE, BERTScore) for similarity to ground-truth. Qualitative human evaluation to assess clarity, coherence, and usefulness (as a potential project extension).

Main Goals

- Develop a systematic benchmarking framework for LLM-powered scientific Discussion generation.
- Identify the most effective LLMs and prompting configurations for each granularity level.

- Explore strategies for optimizing prompts and few-shot examples to improve generation quality.
- Provide insights into the potential and limitations of LLMs in supporting scientific writing tasks.

Relevant Literature

- Docekal, M., Fajcík, M., & Šmrz, P. (2024). *OARelatedWork: A Large-Scale Dataset of Related Work Sections with Full-texts from Open Access Sources*. arXiv:2405.01930.
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- Li, X., & Ouyang, J. (2024). *Related Work and Citation Text Generation: A Survey*. arXiv:2404.11588.
- Agarwal, S., Sahu, G., Puri, A., Laradji, I. H., Dvijotham, K. D. J., Stanley, J., Charlin, L., & Pal, C. (2024). *LitLLMs, LLMs for Literature Review: Are We There Yet?* arXiv:2412.15249.
- Liu, X., Song, R., Wang, X., & Chen, X. (2025). *Select, Read, and Write: A Multi-Agent Framework of Full-Text-based Related Work Generation*. arXiv:2505.19647.
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